9	**(a)**	Distinguish between *gravitational field strength* and *acceleration of free fall*.	

[2]

Solution:

Gravitational field strength is the gravitational force per unit mass acting on a small mass at a point in a gravitational field. **1**

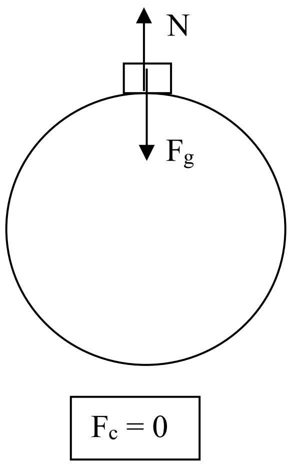
While acceleration of free fall is the change in velocity per unit time (i.e. acceleration) produced by gravitational force alone acting on a mass. **1**

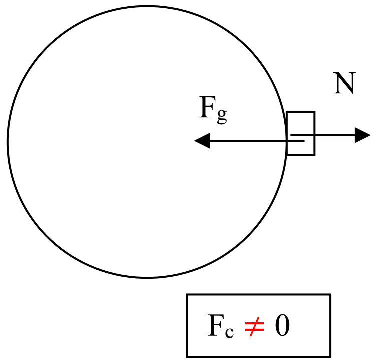
(b) Assuming Earth to be a sphere, explain with the help of free body diagrams, the difference in the weight of a person measured on an electronic balance at the poles and at the equator.

[5]

Pole

Equator





Free Body diagrams with forces identified, and length of vectors at poles are equal, while at the equator F_g vector is longer than N vector.

N: Normal contact force exerted by electronic scale on person
F_g: Gravitational force exerted by Earth on person

1 mark for FBDs awarded if the following are shown:

- *Forces are identified and named*
- *At Poles : correct N and F_g (length and direction)(N and F_g vectors are equal in length)* **1**
- *At Equator : correct N and F_g (length and direction)(F_g vector is longer than N vector)* **1**

Electronic balance reads the apparent weight of the person which is given by the magnitude of N.

At a pole, person does not undergo circular motion, hence centripetal force F_c equals zero. Thus N equals in magnitude to F_g. **1**

At the equator, person undergoes circular motion, hence F_c is non-zero. Thus N is less than F_g. **1**

Since the Earth is assumed to be a perfect sphere, its radius at the poles and equator are equal, hence F_g at both locations are equal. This implies that N at the poles is greater than N at the equator, and hence the apparent weight of the person is lesser at the equator. **1**

(c) Assuming Earth to be a sphere of radius 6.4×10^6 m, calculate the mass of Earth.

mass of Earth = ……………………… kg [2]

Solution:

g = G M/r^2

$g_{\text{at Earth's surface}}$ = G M_E/r_E^2 1

M_E = $g_{\text{at Earth's surface}}$ r_E^2 / G

M_E = 9.81(6.4x10^6)^2 / 6.67 x10^{-11} = 6.02425 x10^{24} = 6.0 x10^{24} kg 1

(d) Calculate the radius of the orbit of a geostationary satellite.

radius = ………………….. m [3]

Solution:

Gravitational force (F_g) exerted by Earth on satellite provides for the centripetal force (F_c) required for the satellite's circular motion. **1**

F_g = F_c

$$\frac{GMm}{r^2} = mr\omega^2 \qquad where\ m\text{: } mass\ of\ satellite$$

$$\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$ **1**

For geostationary orbit, T = 24 h
From (c), $M = 6.02425\ \text{x}10^{24} kg$

$$(24\ x\ 60\ x\ 60)^2 = \frac{4\pi^2 r^3}{(6.67\ x\ 10^{-11})(6.02425\ \text{x}10^{24})}$$

$$r = 4.2\ x\ 10^7\ m$$ **1**

(e) **(i)** A space mission is to launch a space shuttle from Earth. The space shuttle comprise of a land rover to be delivered to the moon.

Using the answer in part **(c)**, calculate the distance from the centre of Earth where the resultant gravitational field strength due to the Earth and the Moon is zero.

The mass of the Moon is 7.4×10^{22} kg and distance between centres of the Earth and the Moon is 3.8×10^{8} m.

distance from centre of Earth = ……………………… m [3]

Solution:

At the null point, $g_{resultant} = 0$ **1**
i.e. $g_E = g_M$

$GM_E/r_E^2 = GM_M/r_M^2$

where r_E is the distance from centre of Earth to null point;
r_M is the distance from centre of the moon to null point

1

$6.02425\text{x } 10^{24} / r_E^{\,2} = 7.4 \times 10^{22} / (3.8 \times 10^{8} - r_E)^2$

$r_E = 3.4209 \times 10^{8} = 3.4 \times 10^{8}$ m **1**

(ii) On Fig. 9.1, sketch the variation of gravitational potential ϕ between the surface of Earth and the surface of the moon.

Include the value obtained in **(e)(i)**.

ϕ ϕ

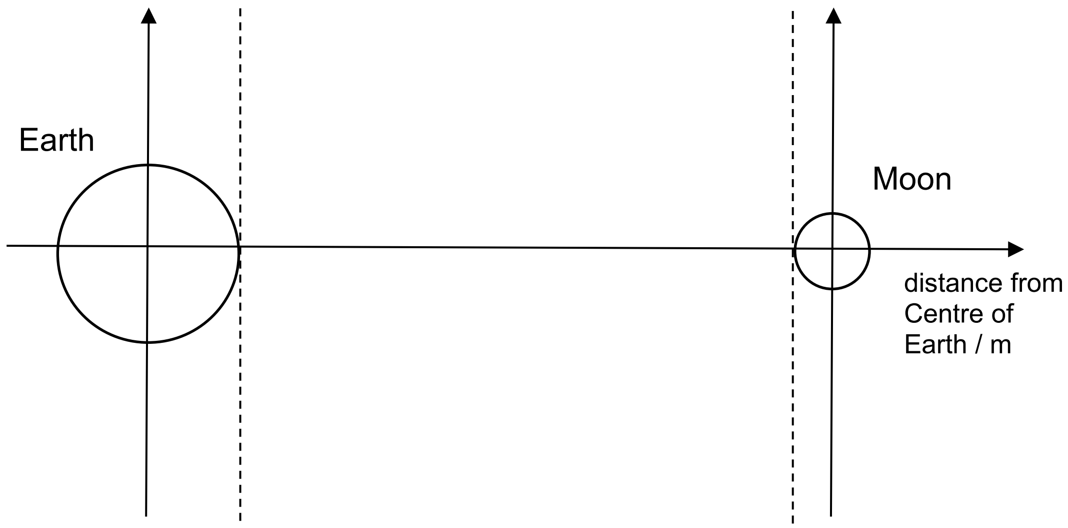


Fig. 9.1

[2]

Solution:

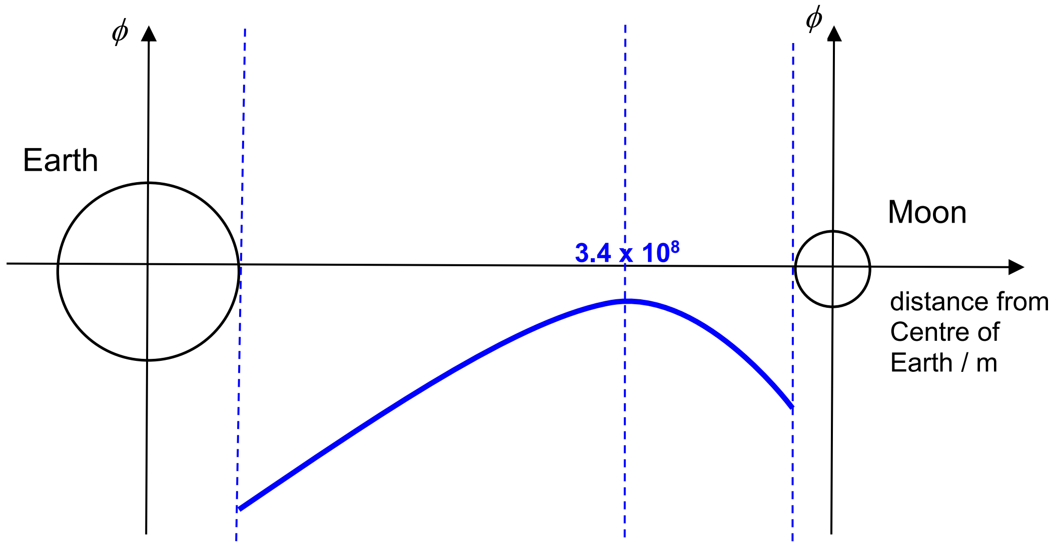


Fig. 9.1

- Maximum point at null point with position labelled 1
- Potential at Earth's surface is more negative than potential at Moon's surface AND asymmetric curve of appropriate shape 1

(iii) Calculate the minimum launch speed of the space shuttle from Earth such that the space shuttle will be able to reach the surface of the moon.

			minimum launch speed = …………………… m s^{-1}	[3]

Solution:

By the Conservation of Energy,
Total Initial Energy $_{\text{Earth surface}}$ = Total Final Energy $_{\text{null point}}$

$$KE_{Earth} + GPE_{Earth} = KE_{null} + GPE_{null}$$

$$\frac{1}{2}mv_{object}{}^{2} + \left(-G\frac{mM_{Earth}}{r_{\text{Earth's centre to Earth's surface}}} - G\frac{mM_{Moon}}{r_{\text{Moon's centre to Earth's surface}}}\right) = 0 + \left(-G\frac{mM_{Earth}}{r_{\text{Earth's centre to Null pt.}}} - G\frac{mM_{Moon}}{r_{\text{Moon's centre to Null pt.}}}\right)$$ **1**

$$\frac{1}{2}v_{object}{}^{2} + \left(-G\frac{6.02425\times10^{24}}{6.4\times10^{6}} - G\frac{6.0\times10^{22}}{(3.8\times10^{8} - 6.4\times10^{6})}\right) = 0 + \left(-G\frac{6.02425\times10^{24}}{3.4209\times10^{8}} - G\frac{6.0\times10^{22}}{(3.8\times10^{8} - 3.4209\times10^{8})}\right)$$ **1**

$$v_{object}{}^{2} = 2G\left(9.4129\times10^{17} + 1.6060\times10^{14} - 1.7610\times10^{16} - 1.5827\times10^{15}\right) = 1.2304\times10^{8}$$

$$v_{object} = 1.1092\times10^{4} = 1.1\times10^{4}$$ **1**

-- END OF PAPER 3 –